

SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 90,000 over 36 months is approved under the aggressive policy (v1.0). Verified monthly income is Rs 39,150 against existing obligations of Rs 2,500; the proposed EMI of Rs 3,076 gives a FOIR of 14.2% (limit 60.0%) and an LTV of 75.0% (limit 90.0%). Risk band: LOW.

For the borrower:

Ravi Kumar, your loan of Rs 90,000 is approved. Your documents agree with each other, and your monthly instalment of Rs 3,076 fits within the affordability limit based on the income we verified.

Key drivers:

- FOIR 14.2% vs limit 60.0%
- LTV 75.0% vs limit 90.0%
- Verified income Rs 39,150 (volatility 4.1%)

Next steps:

- Issue sanction letter and collect e-mandate for the EMI.
- Archive the credit memo PDF with the file.

1. DECISION

Decision: APPROVED Confidence: 0.88

Reason codes: none

1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Ravi Kumar's loan is approved because every policy test passed.

Affordability and collateral are inside the limits: FOIR 14.2% vs limit 60.0% (-45.8 pp under); LTV 75.0% vs limit 90.0% (-15.0 pp under).

No warning was raised on the documents.

The model's risk band is LOW (score 0.2611 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.437336).

What pushes risk up: ltv (loan / asset value) (75.0% vs a typical 68.1%, +0.012); repayment history score (70.0% vs a typical 71.7%, +0.003).

What pulls risk down: foir (obligations / income) (14.2% vs a typical 41.1%, -0.081); emi-to-income ratio (7.9% vs a typical 34.4%, -0.040); employment stability (95.9% vs a typical 71.8%, -0.024).

The risk score is a secondary signal; the policy rules above decide the outcome.

Mathematics:

PASS FOIR 14.2% vs limit 60.0% (-45.8 pp under)

PASS LTV 75.0% vs limit 90.0% (-15.0 pp under)

PASS age 19 vs minimum 18 (+1 years)

PASS verified monthly income Rs 39,150

risk score 0.261123 = base 0.437336 + sum SHAP -0.176213 (method additive_exact, additivity error 0.0)

phi(foir) = -0.0806 [value 14.2% vs baseline 41.1%]

phi(emi_to_income) = -0.0398 [value 7.9% vs baseline 34.4%]

phi(employment_stability) = -0.0241 [value 95.9% vs baseline 71.8%]

phi(income_volatility) = -0.0183 [value 4.1% vs baseline 22.4%]

phi(income_mismatch_percentage) = -0.0124 [value 0.3% vs baseline 12.7%]

2. BORROWER SUMMARY (declared facts)

ref_id: REF-APP-AGE-009

full_name: Ravi Kumar

age: 19

location_tier: tier_2

employment_type: gig_platform

declared_monthly_income: 39,000.00

loan_request.amount: 90,000.00

loan_request.down_payment: 30,000.00

loan_request.tenure_months: 36

loan_request.vehicle_price: 120,000.00

3. EXTRACTED FACTS (evidence with citations)

full_name = Ravi Kumar conf 0.95 [application_form p.None] auto
monthly_income = 39,000.00 conf 0.95 [application_form p.None] auto
loan_amount = 90,000.00 conf 0.95 [application_form p.None] auto
vehicle_price = 120,000.00 conf 0.95 [application_form p.None] auto
full_name = Ravi Kumar conf 0.85 [kyc.txt p.1] auto
identity = XXXX-XXXX-4321 conf 0.9 [kyc.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [kyc.txt p.1] auto
monthly_income = 38,900.00 conf 0.92 [bank_statement.txt p.1] auto
bank_credits = [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0] conf 0.9 [bank_statement.txt p.1] auto
existing_monthly_obligations = 2,500.00 conf 0.92 [bank_statement.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [bank_statement.txt p.1] auto
vehicle_price = 120,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
loan_amount = 90,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
platform_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto
monthly_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto

4. CALCULATED VALUES (deterministic engine, formula v1)

verified_income: 39,150.00
existing_obligations: 2,500.00
emi: 3,075.99
foir: 0.1424
ltv: 0.75
income_volatility: 0.0408
income_stability: 0.9592
loan_to_income: 0.1916
emi_to_income: 0.0786
input.annual_rate: 14.00
input.existing_monthly_obligations: 2,500.00
input.loan_amount: 90,000.00
input.monthly_incomes: [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0]
input.tenure_months: 36
input.vehicle_price: 120,000.00

5. RISK SIGNAL (secondary, never decides)

score 0.2611 band LOW model xgb-v1/fallback

6. AI REASONING

strengths: FOIR 14% is comfortably serviceable
strengths: LTV 75% leaves borrower equity cushion
risks: INCOME_MISMATCH: Monthly income differs across documents by 0.3%
reasoning: Deterministic FOIR 14.24%, LTV 75.00%; XGBoost band LOW (score 0.2611).

7. POLICY

profile aggressive version 1.0
limit.foir_max: 0.6
limit.ltv_max: 0.9
limit.max_tenure_months: 72
limit.min_age: 18

8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.